# Arista Access Node MAR-ACC-04 Marathahalli Core BGP Route Churn and OSPF Adjacency Loss SOP

**Node:** MAR-ACC-04 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8631

## Description

This SOP addresses the failure domain of routing protocol flap in the Access Node MAR-ACC-04 located at Marathahalli Core, which is running Arista network equipment. The failure is characterized by BGP route churn and OSPF adjacency loss, resulting in the ALARM_8631 major alarm. The root cause of this failure is attributed to the complex interaction between BGP and OSPF routing protocols, which can lead to oscillations and instability in the control plane. This can be exacerbated by factors such as misconfigured peers, inadequate network design, or hardware issues. In the case of Arista equipment, the issue may be further complicated by the use of features such as Fast Convergence and LACP, which can amplify the effects of protocol instability. The diagnostic and resolution steps outlined in this SOP are designed to identify and rectify the root cause of the failure, ensuring the stability and reliability of the network.

## Pre-requisite Checks

- Pre-requisite: Verify the site location and confirm that the failure is indeed occurring on Node ID MAR-ACC-04 at Marathahalli Core

- Pre-requisite: Check the status of the redundant links and confirm that they are functional

- Pre-requisite: Verify that the network design and configuration are adequate and not contributing to the failure

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfigured BGP peers | High |
| Inadequate network design | Medium |
| Hardware issues | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show ip bgp neighbors, Expected Output: BGP neighbor is 10.10.10.10, remote AS 65001, local AS 65002, neighbor version 4, neighbor sent and received BGP routes, Description: This command verifies the BGP neighbor status and helps identify any misconfigurations

Command: show ospf neighbors, Expected Output: OSPF Neighbor is 10.10.10.10, remote IP 10.10.10.10, local IP 10.10.10.11, Description: This command verifies the OSPF neighbor status and helps identify any misconfigurations
```

```
Command: show lldp neighbors, Expected Output: LLDP neighbor is 10.10.10.10, remote port 1, local port 2, Description: This command verifies the LLDP neighbor status and helps identify any physical layer issues
```

## Resolution Steps

Step 1: Step: Verify and correct BGP peer configuration, Command: conf t router bgp 65002 neighbor 10.10.10.10 remote-as 65001 neighbor 10.10.10.10 timers 10 20 30 exit , Description: This step involves verifying the BGP peer configuration and correcting any misconfigurations. The command updates the BGP neighbor configuration with the correct timers.

Step 2: Step: Verify and correct OSPF neighbor configuration, Command: conf t router ospf 1 neighbor 10.10.10.10 priority 100 neighbor 10.10.10.10 dead-interval 30 exit , Description: This step involves verifying the OSPF neighbor configuration and correcting any misconfigurations. The command updates the OSPF neighbor configuration with the correct priority and dead interval.

Step 3: Step: Verify and correct LLDP neighbor configuration, Command: conf t interface Ethernet1 lldp transmit lldp receive lldp timer 10 lldp holdtime 30 lldp reinit exit , Description: This step involves verifying the LLDP neighbor configuration and correcting any misconfigurations. The command updates the LLDP neighbor configuration with the correct timers.

## Post-Check Verification

```
Command: show ip bgp neighbors, Expected Output: BGP neighbor is 10.10.10.10, remote AS 65001, local AS 65002, neighbor version 4, neighbor sent and received BGP routes, Description: This command verifies the BGP neighbor status and confirms that the correction was successful

Command: show ospf neighbors, Expected Output: OSPF Neighbor is 10.10.10.10, remote IP 10.10.10.10, local IP 10.10.10.11, Description: This command verifies the OSPF neighbor status and confirms that the correction was successful

Command: show lldp neighbors, Expected Output: LLDP neighbor is 10.10.10.10, remote port 1, local port 2, Description: This command verifies the LLDP neighbor status and confirms that the correction was successful
```

## Roll-back Procedures

- Step: Restore the original BGP peer configuration, Command: conf t router bgp 65002 no neighbor 10.10.10.10 remote-as 65001 no neighbor 10.10.10.10 timers 10 20 30 exit , Description: This step involves restoring the original BGP peer configuration in case the correction failed.

- Step: Restore the original OSPF neighbor configuration, Command: conf t router ospf 1 no neighbor 10.10.10.10 priority 100 no neighbor 10.10.10.10 dead-interval 30 exit , Description: This step involves restoring the original OSPF neighbor configuration in case the correction failed.

- Step: Restore the original LLDP neighbor configuration, Command: conf t interface Ethernet1 no lldp transmit no lldp receive no lldp timer 10 no lldp holdtime 30 lldp reinit exit , Description: This step involves restoring the original LLDP neighbor configuration in case the correction failed.